

Statistical Mechanics

Statistical Distribution

This determines the most probable way in which a certain total amount of energy 'E' is distributed among the 'N' members of a system of particles in thermal equilibrium at absolute temperature, T.

Thus Statistical Mechanics reflects overall behavior of system of many particles.

Suppose $n(\epsilon)$ is the no. of particles having energy, ϵ

then
$$n(\epsilon) = g(\epsilon)f(\epsilon)$$

where $g(\epsilon)$ = no. of states of energy ϵ or statistical weight corresponding to energy ϵ

$f(\epsilon)$ = distribution function

= average no. of particles in each state of energy ϵ

= probability of occupancy of each state of energy ϵ

Statistical Distribution are of three kinds

1. Identical particles that are sufficiently far apart to be distinguishable.

Example: molecules of a gas.

Negligible overlapping of ψ

Maxwell-Boltzmann Statistics

2. Indistinguishable identical particles of '0' or integral spin.

Example: Bosons (don't obey the exclusion principle)

Overlapping of ψ

Bose-Einstein Statistics

3. Indistinguishable identical particles with odd-half integral spin ($1/2, 3/2, 5/2 \dots$).

Example: Fermions (obey the exclusion principle)

Fermi-Dirac Statistics

Maxwell-Boltzmann Statistics

(Classical Approach)

According to this law number of identical and distinguishable particles in a system at temperature, T having energy ϵ is

$n(\epsilon) = (\text{No. of states of energy } \epsilon) \cdot (\text{average no. of particles in a}$

$$n(\epsilon) = g(\epsilon) \cdot A e^{-\epsilon/kT} \quad \text{state of energy } \epsilon \quad (i)$$

Here A is a constant and $f_{M.B.}(\epsilon) = A e^{-\epsilon/kT}$

Equation (i) represents the Maxwell-Boltzmann Distribution Law

This law cannot explain the behavior of photons (Black body radiation) or of electrons in metals (specific heat, conductivity)

Applications of M.B. Statistics

(i) Molecular energies in ideal gas

No. of molecules having energies between ϵ and $\epsilon + d\epsilon$ is given by

$$n(\epsilon)d\epsilon = \{g(\epsilon)d\epsilon\} \{f(\epsilon)\}$$

$$n(\epsilon) = Ag(\epsilon)e^{-\epsilon/kT}d\epsilon \quad (i)$$

The momentum of a molecule having energy ϵ is given by

$$p = \sqrt{2m\epsilon} \quad \Rightarrow \quad dp = \frac{m d\epsilon}{\sqrt{2m\epsilon}}$$

No. of states in momentum space having momentum between p and $p + dp$ is proportional to the volume element i.e.

$$\alpha \iiint dp_x dp_y dp_z$$

$$\propto \frac{4}{3}\pi(p+dp)^3 - \frac{4}{3}\pi p^3$$

$$\propto 4\pi p^2 dp$$

$$g(p)dp \propto 4\pi p^2 dp$$

$$g(p)dp = Bp^2 dp \quad \text{B is a constant}$$

Each momentum corresponds to a single ϵ

$$g(\epsilon)d\epsilon = Bp^2 dp$$

$$g(\epsilon)d\epsilon = B \cdot 2m\epsilon \cdot \frac{m d\epsilon}{\sqrt{2m\epsilon}}$$

$$g(\epsilon)d\epsilon = 2m^{3/2}B\sqrt{\epsilon}d\epsilon$$

Put in (i)

No. of molecules with energy between ϵ and $\epsilon + d\epsilon$

$$n(\epsilon)d\epsilon = 2m^{3/2}BA\sqrt{\epsilon}e^{-\epsilon/kT}$$

$$n(\epsilon)d\epsilon = C\sqrt{\epsilon}e^{-\epsilon/kT} \quad (\text{ii})$$

Total No. of molecules is N

$$N = \int_0^{\infty} n(\epsilon)d\epsilon = C \int_0^{\infty} \sqrt{\epsilon}e^{-\epsilon/kT} d\epsilon$$

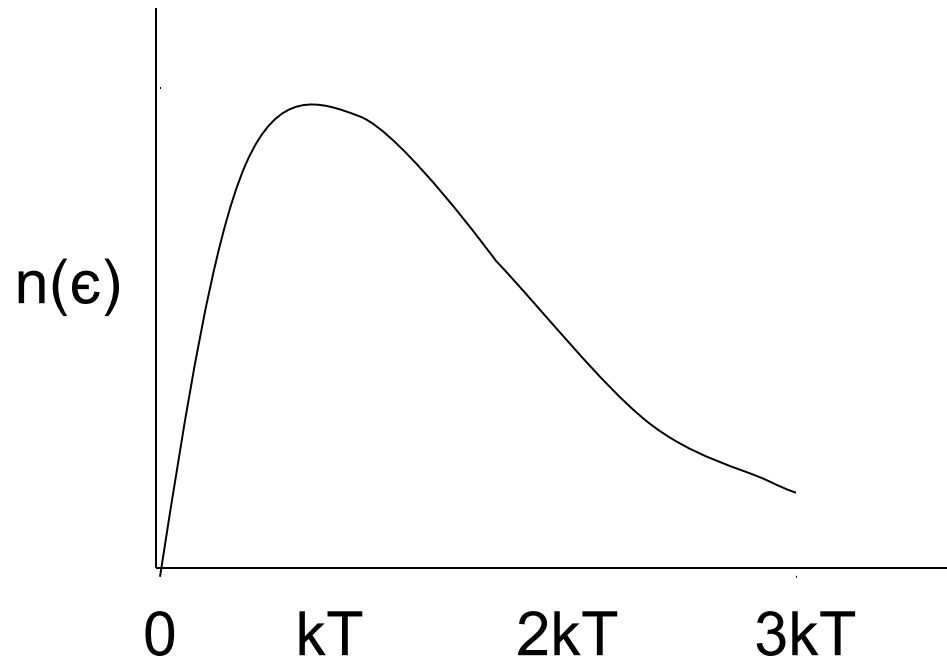
Using identity $\int_0^{\infty} x^{1/2} e^{-ax} dx = \frac{1}{2a} \sqrt{\frac{\pi}{a}}$

$$C = \frac{2\pi N}{(\pi kT)^{3/2}}$$

Put value of C in equation (ii)

$$n(\epsilon)d\epsilon = \frac{2\pi N}{(\pi kT)^{3/2}} \sqrt{\epsilon} e^{-\epsilon/kT} d\epsilon$$

This equation gives the number of molecules with energies between ϵ and $\epsilon + d\epsilon$ in a sample of an ideal gas that contains N molecules and whose absolute temperature is T. It is called molecular energy distribution equation.



Maxwell-Boltzmann energy distribution curve for the molecules of an ideal gas.

Average Molecular Energy

Total energy of the system $E = \int_0^{\infty} \epsilon n(\epsilon) d\epsilon$

$$E = \frac{2\pi N}{(\pi kT)^{3/2}} \int_0^{\infty} \epsilon^{3/2} e^{-\epsilon/kT} d\epsilon$$

Using

$$\int_0^{\infty} x^{3/2} e^{-ax} dx = \frac{3}{4a^2} \sqrt{\frac{\pi}{a}}$$

$$E = \frac{2\pi N}{(\pi kT)^{3/2}} \left[\frac{3}{4} (kT)^2 \sqrt{\pi kT} \right]$$

$$E = \frac{3}{2} NkT$$

$$\text{Average energy per molecule} = \bar{\mathcal{E}} = \frac{E}{N}$$

$$\bar{\mathcal{E}} = \frac{3}{2} kT$$

This is the average molecular energy and is independent of the molecular mass

At Room Temperature its value is 0.04 eV.

Equipartition of Energy: Equipartition Theorem

Average Kinetic Energy associated with each degree of freedom of a molecule is

$$\frac{1}{2}kT$$

Distribution of molecular speed

$$K.E. \Rightarrow \varepsilon = \frac{1}{2}mv^2$$
$$\Rightarrow d\varepsilon = mv dv$$

Put this in molecular energy distribution equation

$$n(v)dv = 4\pi N \left(\frac{m}{2\pi kT} \right)^{3/2} v^2 e^{-mv^2/2kT} dv$$

It is called molecular speed distribution formula.

RMS Speed

$$v_{rms} = \sqrt{\overline{v^2}}$$

It is the square root of the average squared molecular speed

$$\overline{\varepsilon} = \frac{1}{2} m \overline{v^2} = \frac{3}{2} kT$$

$$v_{rms} = \sqrt{\frac{3kT}{m}}$$

Average Speed

$$\bar{v} = \frac{1}{N} \int_0^{\infty} v n(v) dv$$

$$= 4\pi \left(\frac{m}{2\pi kT} \right)^{3/2} \int_0^{\infty} v^3 e^{-mv^2/2kT} dv$$

Using

$$\int_0^{\infty} x^3 e^{-ax^2} dx = \frac{1}{2a^2}$$

$$\bar{v} = 4\pi \left(\frac{m}{2\pi kT} \right)^{3/2} \cdot \left(\frac{4k^2 T^2}{2m^2} \right) = \sqrt{\frac{8kT}{\pi m}}$$

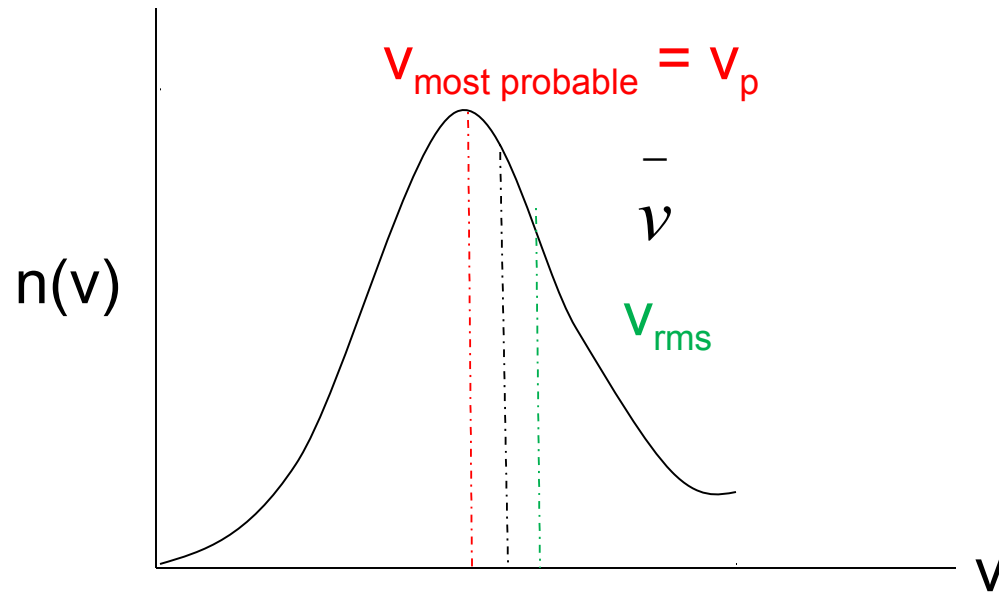
Most Probable Speed

$$n(v) = 4\pi N \left(\frac{m}{2\pi kT} \right)^{3/2} v^2 e^{-mv^2/2kT}$$

$$\frac{dn(v)}{dv} = 4\pi N \left(\frac{m}{2\pi kT} \right)^{3/2} \frac{d}{dv} \left[v^2 e^{-mv^2/2kT} \right] = 0$$

$$4\pi N \left(\frac{m}{2\pi kT} \right)^{3/2} \neq 0$$

$$\frac{d}{dv} \left[v^2 e^{-mv^2/2kT} \right] = 0 \quad \Rightarrow \quad v_p = \sqrt{\frac{2kT}{m}}$$



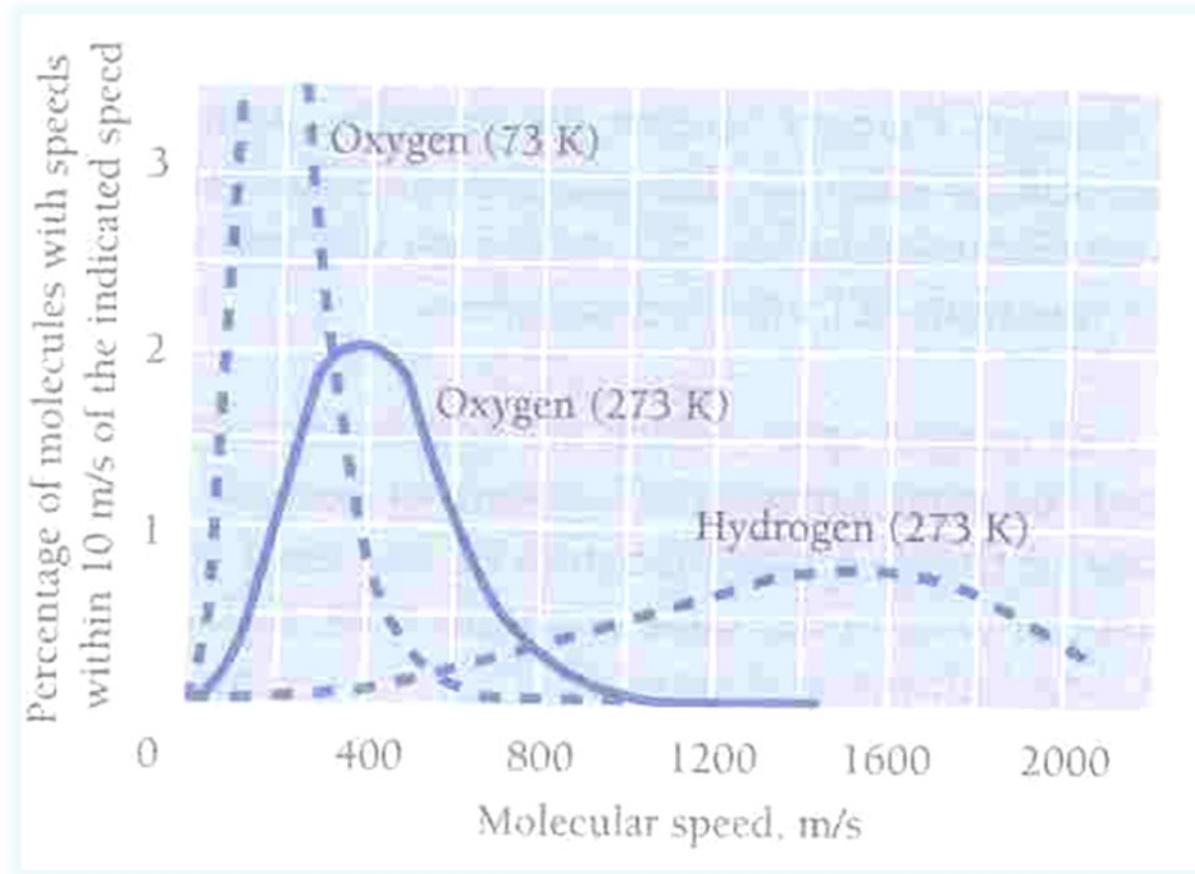
M.B. Speed Distribution Curve

(i) Speed distribution is not symmetrical

(ii)
$$v_p < \bar{v} < v_{rms}$$

$$\sqrt{\frac{2kT}{m}} < \sqrt{\frac{8kT}{\pi m}} < \sqrt{\frac{3kT}{m}}$$

Variation in molecular speed with 'T' and ' molecular mass'



This curve suggests that most probable speed $\propto \sqrt{T}$

Also the most probable speed $\propto 1/\sqrt{m}$

Quantum Statistics (Indistinguishable identical particles)

Bosons

1. '0' or integral spin
2. Do not obey exclusion principle
3. Symmetric wave function
4. Any number of bosons can exist in the same quantum state of the system

Fermions

1. Odd half integral spin
2. Obey exclusion principle
3. Anti-symmetric wave function
4. Only one fermion can exist in a particular quantum state of the system

Consider a system of two particles, 1 and 2, one of which is in state 'a' and the other in state 'b'.

When particles are distinguishable then

$$\psi' = \psi_a(1)\psi_b(2)$$

$$\psi'' = \psi_a(2)\psi_b(1)$$

When particles are indistinguishable then

$$\text{For Bosons, } \psi_B = \frac{1}{\sqrt{2}} [\psi_a(1)\psi_b(2) + \psi_a(2)\psi_b(1)]$$

symmetric

$$\text{For Fermions, } \psi_F = \frac{1}{\sqrt{2}} [\psi_a(1)\psi_b(2) - \psi_a(2)\psi_b(1)]$$

anti-symmetric

ψ_F becomes zero when 'a' is replaced with 'b' i.e.

$$\psi_F = \frac{1}{\sqrt{2}} [\psi_a(1)\psi_a(2) - \psi_a(2)\psi_a(1)] = 0$$

Thus two Fermions can't exist in same state (Obey Exclusion Principle)

Bose-Einstein Statistics

Any number of particles can exist in one quantum state.

Distribution function can be given by

$$f_{B.E.}(\epsilon) = \frac{1}{e^{\alpha} e^{\epsilon/kT} - 1}$$

where α may be a function of temperature, T.

For Photon gas $\alpha = 0 \Rightarrow e^{\alpha} = 1$

-1 in denominator indicates multiple occupancy of an energy state by Bosons

Bose-Einstein Condensation

If the temperature of any gas is reduced, the wave packets grow larger as the atoms lose momentum according to uncertainty principle.

When the gas becomes very cold, the dimensions of the wave packets exceeds the average atomic spacing resulting into overlapping of the wave packets. If the atoms are bosons, eventually, all the atoms fall into the lowest possible energy state resulting into a single wave packet. This is called Bose-Einstein condensation.

The atoms in such a Bose-Einstein condensate are barely moving, are indistinguishable, and form one entity – a superatom.

Fermi-Dirac Statistics

Obey Pauli's exclusion Principle

Distribution function can be given by

$$f_{F.D.}(\epsilon) = \frac{1}{e^{\alpha} e^{\epsilon/kT} + 1}$$

$f(\epsilon)$ can never exceed 1, whatever be the value of α , ϵ and T . So only one particle can exist in one quantum state.

α is given by

$$\alpha = -\frac{\epsilon_F}{kT}$$

where ϵ_F is the Fermi Energy

then

$$f_{F.D.}(\epsilon) = \frac{1}{e^{(\epsilon - \epsilon_F)/kT} + 1}$$

Case I:

$$T = 0$$

$$(\epsilon - \epsilon_F)/kT = -\infty \quad \text{For } \epsilon < \epsilon_F$$

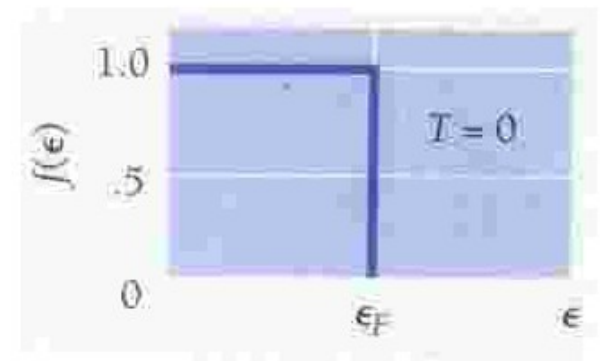
$$(\epsilon - \epsilon_F)/kT = +\infty \quad \text{For } \epsilon > \epsilon_F$$

For $\epsilon < \epsilon_F$

$$f_{F.D.}(\epsilon) = \frac{1}{e^{-\infty} + 1} = 1$$

For $\epsilon > \epsilon_F$

$$f_{F.D.}(\epsilon) = \frac{1}{e^{+\infty} + 1} = 0$$



Thus at $T = 0$, For $\epsilon < \epsilon_F$ all energy states from $\epsilon = 0$ to ϵ_F are occupied as

$$f_{F.D.}(\epsilon) = 1$$

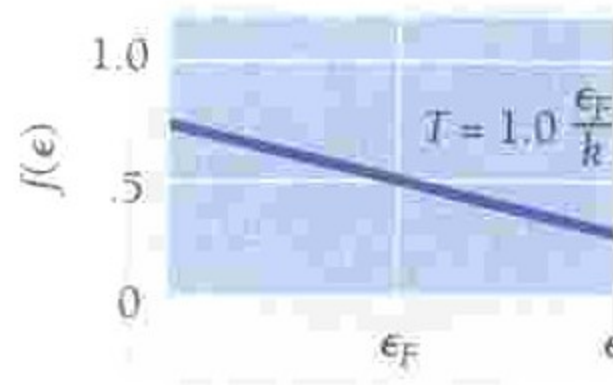
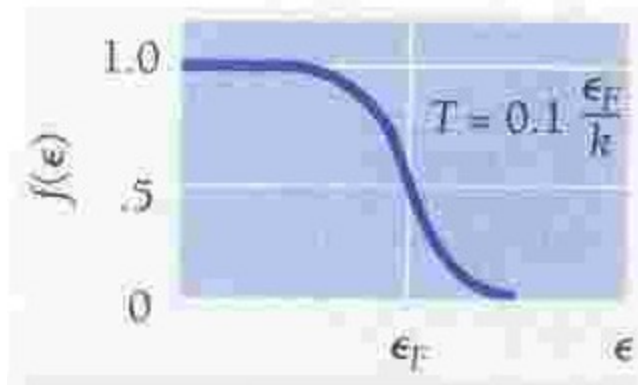
Thus at $T = 0$, all energy states for which $\epsilon > \epsilon_F$ are vacant.

$$f_{F.D.}(\epsilon) = 0$$

Case II:

$$T > 0$$

Some of the filled states just below ϵ_F becomes vacant while some just above ϵ_F become occupied.

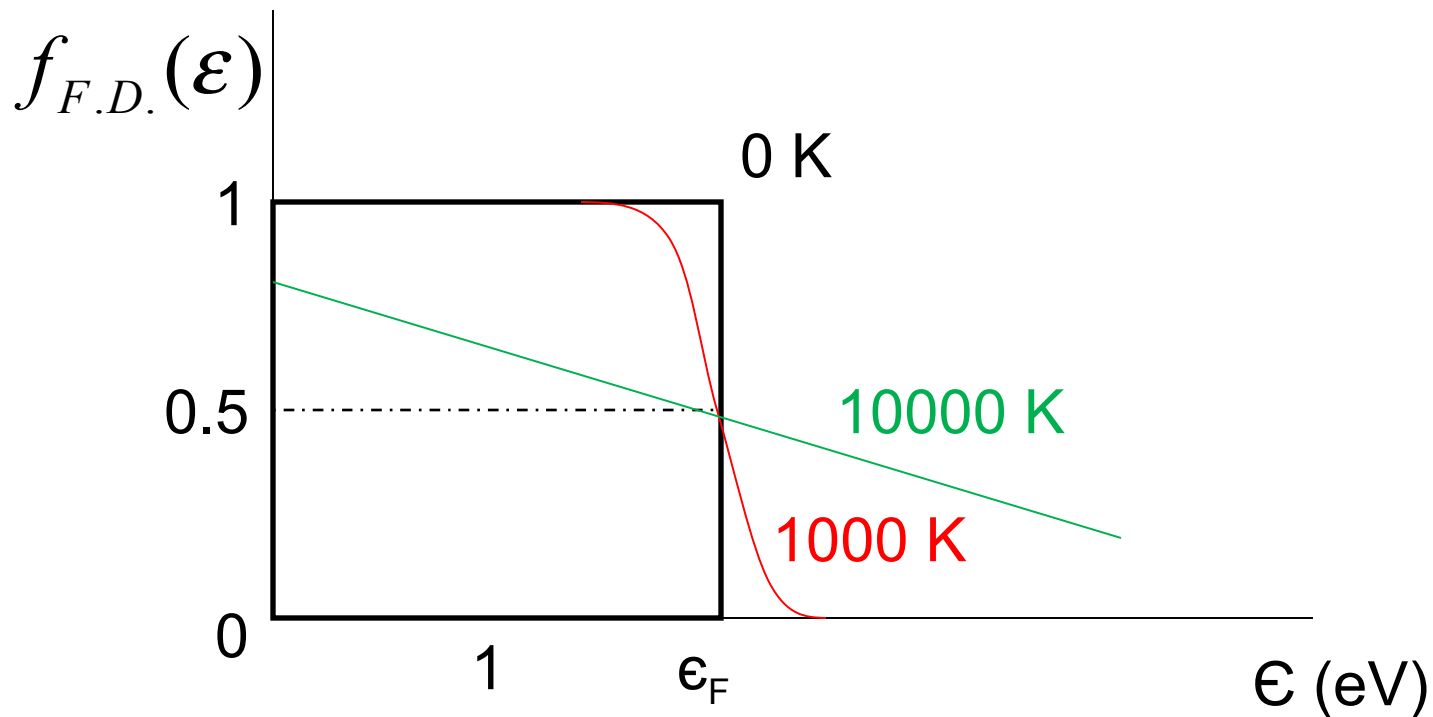


Case III:

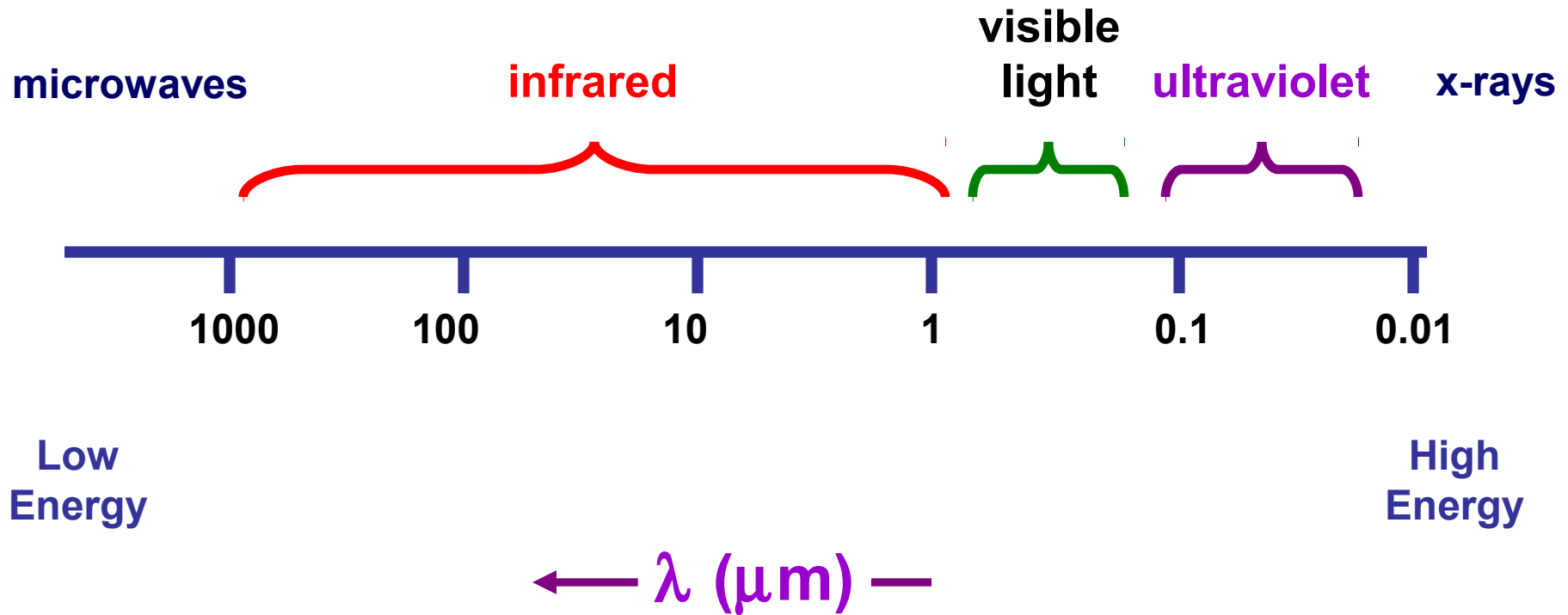
At $\epsilon = \epsilon_F$

$$f_{F.D.}(\epsilon) = \frac{1}{e^0 + 1} = \frac{1}{2} \quad \text{For all } T$$

Probability of finding a fermion (i.e. electron in metal) having energy equal to fermi energy (ϵ_F) is $\frac{1}{2}$ at any temperature.



Electromagnetic Spectrum



Black Body Radiation

All objects radiate electromagnetic energy continuously regardless of their temperatures.

Though which frequencies predominate depends on the temperature.

At room temperature most of the radiation is in infrared part of the spectrum and hence is invisible.

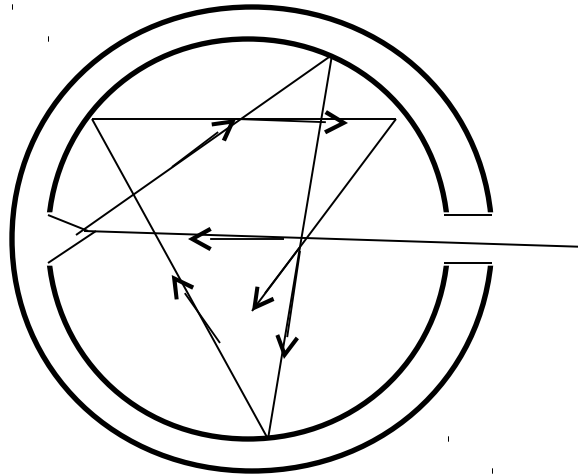
The ability of a body to radiate is closely related to its ability to absorb radiation.

A body at a constant temperature is in thermal equilibrium with its surroundings and must absorb energy from them at the same rate as it emits energy.

A perfectly **black body** is the one which absorbs completely all the radiation, of whatever wavelength, incident on it.

Since it neither transmits any radiation, it appears black whatever the color of the incident radiation may be.

There is no surface available in practice which will absorb all the radiation falling on it.



The cavity walls are constantly emitting and absorbing radiation, and this radiation is known as black body radiation.

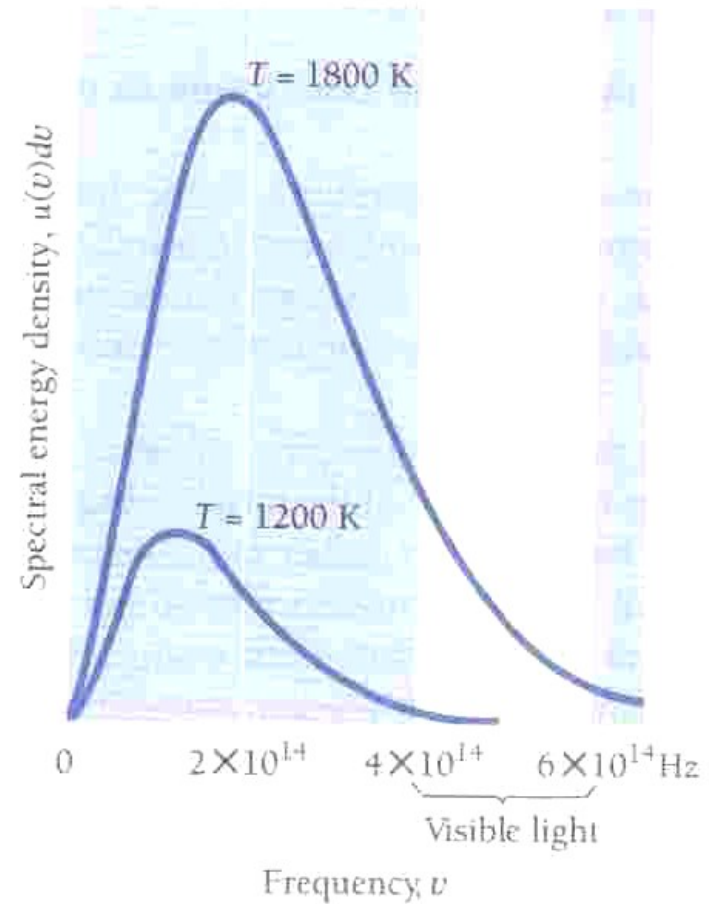
Characteristics of Black Body Radiation

(i) The total energy emitted per second per unit area (radiancy E or area under curve) increases rapidly with increasing temperature.

(ii) At a particular temperature, the spectral radiance is maximum at a particular frequency.

(iii) The frequency (or wavelength, λ_m) for maximum spectral radiance decreases in direct proportion to the increase in temperature. This is called “Wien’s displacement law”

$$\lambda_m \times T = \text{constant } (2.898 \times 10^{-3} \text{ m.K})$$



Planck's Radiation Law

Planck assumed that the atoms of the walls of cavity radiator behave as oscillators with energy

$$\mathcal{E}_n = nh\nu \quad n = 0, 1, 2, 3, \dots$$

The average energy of an oscillator is

$$\bar{\mathcal{E}} = \frac{\mathcal{E}}{N} \quad \text{N is total no. of Oscillators}$$

$$\bar{\mathcal{E}} = \frac{h\nu}{e^{h\nu/kT} - 1} \quad (i)$$

Thus the energy density (u_ν) of radiation in the frequency range ν to $\nu + d\nu$ is

$$u_\nu d\nu = \frac{8\pi\nu^2 d\nu}{c^3} \times \bar{\mathcal{E}}$$

$$u_\nu d\nu = \frac{8\pi\nu^2 d\nu}{c^3} \left(\frac{h\nu}{e^{h\nu/kT} - 1} \right)$$

$$u_\nu d\nu = \frac{8\pi h\nu^3}{c^3} \frac{d\nu}{e^{h\nu/kT} - 1}$$

This is Planck's Radiation formula in terms of frequency.

In terms of wavelength

$$u_\lambda d\lambda = \frac{8\pi hc}{\lambda^5} \frac{d\lambda}{e^{hc/\lambda kT} - 1}$$

This is Planck's Radiation formula in terms of wavelength.

Case I : (Rayleigh-Jeans Law)

$$u_{\lambda} d\lambda = \frac{8\pi hc}{\lambda^5} \frac{d\lambda}{e^{hc/\lambda kT} - 1}$$

When λ is large then $e^{hc/\lambda kT} \approx 1 + \frac{hc}{\lambda kT}$

$$u_{\lambda} d\lambda = \frac{8\pi hc}{\lambda^5 \left(1 + \frac{hc}{\lambda kT} - 1 \right)} d\lambda$$

$$u_{\lambda} d\lambda = \frac{8\pi kT}{\lambda^4} d\lambda$$

This is Rayleigh-Jeans Law for longer λ 's.

Case II : (Wein's Law)

$$u_{\lambda} d\lambda = \frac{8\pi hc}{\lambda^5} \frac{d\lambda}{e^{hc/\lambda kT} - 1}$$

When λ is very small then $e^{hc/\lambda kT} \gg 1$

$$u_{\lambda} d\lambda = \frac{8\pi hc}{\lambda^5} e^{-hc/\lambda kT} d\lambda$$

This is Wein's Law for small λ 's.

$$u_{\lambda} d\lambda = \frac{A}{\lambda^5} e^{B/\lambda kT} d\lambda$$

This is another form of Wein's Law..

Here A and B are constants.

Stefan's Law

Planck's Radiation formula in terms of frequency is

$$u_{\nu} d\nu = \frac{8\pi h \nu^3}{c^3} \frac{d\nu}{e^{h\nu/kT} - 1}$$

$$\frac{c}{4} u_{\nu} d\nu = \frac{2\pi h \nu^3}{c^2} \frac{d\nu}{e^{h\nu/kT} - 1}$$

The spectral radiance E_{ν} is related to the energy density u_{ν} by

$$E_{\nu} = \frac{c}{4} u_{\nu}$$

$$E_{\nu} d\nu = \frac{2\pi h}{c^2} \frac{\nu^3 d\nu}{e^{h\nu/kT} - 1}$$

$$E = \int_0^{\infty} E_{\nu} d\nu = \frac{2\pi h}{c^2} \int_0^{\infty} \frac{\nu^3 d\nu}{e^{h\nu/kT} - 1}$$

Let $\frac{h\nu}{kT} = x \quad \Rightarrow \quad \nu = \frac{kT}{h} x \quad \text{and} \quad d\nu = \frac{kT}{h} dx$

$$E = \frac{2\pi k^4 T^4}{h^3 c^2} \int_0^{\infty} \frac{x^3 dx}{e^x - 1}$$

But $\int_0^{\infty} \frac{x^3 dx}{e^x - 1} = \frac{\pi^4}{15}$

$$E = \frac{2\pi^5 k^4}{15h^3 c^2} T^4$$

Let $\frac{2\pi^5 k^4}{15h^3 c^2} = \sigma$ (Stefan's constant)

$$E = \sigma T^4$$

This is Stefan's Law.

Here $\sigma = \frac{2\pi^5 k^4}{15h^3 c^2} = 5.67 \times 10^{-5} \text{ erg}/(\text{cm}^2 \cdot \text{sec} \cdot \text{K}^4)$
 $= 5.67 \times 10^{-8} \text{ W}/(\text{m}^2 \cdot \text{K}^4)$

Wein's Displacement Law

Planck's Radiation formula in terms of frequency is

$$u_{\lambda} = \frac{8\pi hc}{\lambda^5} \frac{1}{e^{hc/\lambda kT} - 1}$$

Using $\frac{d(u_{\lambda})}{d\lambda} = 0$ for $\lambda = \lambda_{\max}$

$$\frac{hc}{kT\lambda_{\max}} = \text{Constant}$$

$$\lambda_m \times T = \text{constant } (2.898 \times 10^{-3} \text{ m.K})$$

Peaks in Black Body radiation shifts to shorter wavelength with increase in temperature.

Specific Heat of Solids

Atoms in solid behave as oscillators.

In case of solids total average energy per atom per degree of freedom is kT ($0.5kT$ from K.E. and $0.5kT$ from P.E.).

Each atom in the solid should have total energy = $3kT$ (3 degrees of freedom).

For one mole of solid, total energy, $E = 3N_0kT$ (classically)

Here N_0 is the Avogadro number.

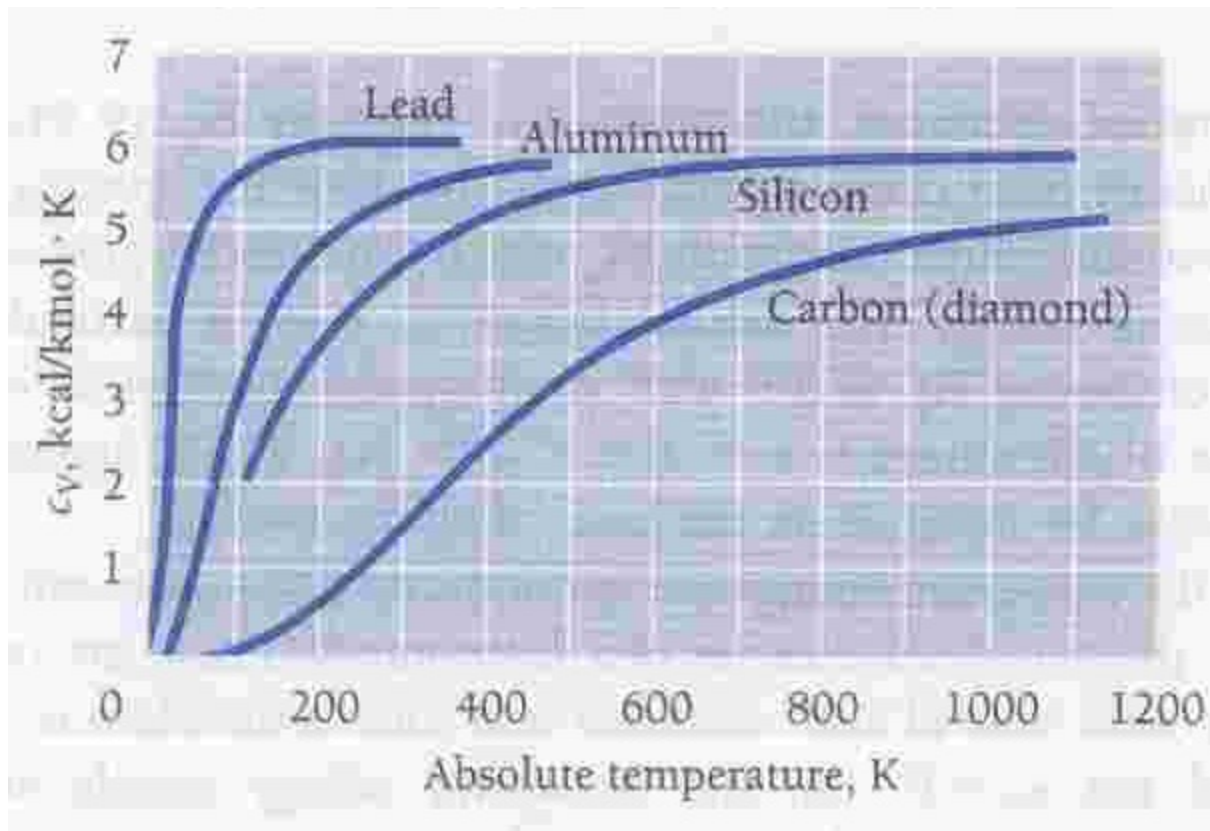
$$E = 3RT$$

Specific heat at constant volume is

$$C_v = \left(\frac{\partial E}{\partial T} \right)_V = 3R \quad \sim 6 \text{ kcal/kmol.K}$$

This is Dulong & Petit's Law.

This means that atomic specific heat (at constant volume) for all solids is approx 6 kcal/kmol.K and is independent of T.



Dulong and Petit's Law fails for light elements such as B, Be and C.

Also it is not applicable at low temperatures for all solids.

Einstein's theory of Specific Heat of Solids

According to it the motion of the atoms in a solid is oscillatory.

The average energy per oscillator atom) is

$$\bar{\epsilon} = \frac{h\nu}{e^{h\nu/kT} - 1}$$

The total energy for one mole of solid in three degrees of freedom.

$$E = 3N_o \frac{h\nu}{e^{h\nu/kT} - 1}$$

The molar specific heat of solid is $C_v = \left(\frac{\partial E}{\partial T} \right)_v$

$$C_v = 3N_o hv \left(\frac{hv}{kT^2} \right) \frac{e^{hv/kT}}{(e^{hv/kT} - 1)^2} = 3N_o k \left(\frac{hv}{kT} \right)^2 \frac{e^{hv/kT}}{(e^{hv/kT} - 1)^2}$$

$$C_v = 3R \left(\frac{hv}{kT} \right)^2 \frac{e^{hv/kT}}{(e^{hv/kT} - 1)^2}$$

This is Einstein's specific heat formula.

Case I:

At high T, $kT \gg hv$ then

$$e^{hv/kT} \approx 1 + \frac{hv}{kT}$$

$$C_v \approx 3R \left(\frac{h\nu}{kT} \right)^2 \frac{\left(1 + \frac{h\nu}{kT} \right)}{\left(1 + \frac{h\nu}{kT} - 1 \right)^2}$$

$$C_v \approx 3R \left(1 + \frac{h\nu}{kT} \right)$$

$$C_v \approx 3R \quad (\because kT \gg h\nu)$$

This is in agreement with Dulong & Petit's Law at high T.

Case II:

At low T, $h\nu \gg kT$ then $e^{h\nu/kT} \gg 1$

$$C_v \approx 3R \left(\frac{h\nu}{kT} \right)^2 e^{-h\nu/kT}$$

This implies that as $T \rightarrow 0, C_v \rightarrow 0$

i.e. is in agreement with experimental results at low T.

Free electrons in a metal

Typically, one metal atom gives one electron.

One mole atoms gives one mole of free electrons, (N_o)

If each free electron can behave like molecules of an ideal gas.
Then

Average K.E. for 1 mole of electron gas, $E_e = \frac{3}{2} N_o kT = \frac{3}{2} RT$

Molar specific heat of electron gas, $(C_v)_e = \left(\frac{\partial E_e}{\partial T} \right)_V = \frac{3}{2} R$

Then total specific heat in metals at high T should be

$$C_v = 3R + \frac{3}{2} R = \frac{9}{2} R$$

But experimentally at high T $C_v \approx 3R$

⇒ Free electrons don't contribute in specific heat,
Why?

Electrons are fermions and have upper limit on the occupancy of the quantum state.

By definition highest state of energy to be filled by a free electron at $T = 0$ is obtained at $\epsilon = \epsilon_F$

The no. of electrons having energy ϵ is

$$N = \int_0^{\epsilon_F} g(\epsilon) d\epsilon \quad \text{where} \quad g(\epsilon) d\epsilon = \frac{8\sqrt{2}\pi V m^{3/2}}{h^3} \sqrt{\epsilon} d\epsilon$$

Here V is the volume of the metal

$$N = \frac{8\sqrt{2}\pi V m^{3/2}}{h^3} \int_0^{\epsilon_F} \sqrt{\epsilon} d\epsilon = \frac{16\sqrt{2}\pi V m^{3/2}}{3h^3} \epsilon_F^{3/2}$$

$$\Rightarrow \epsilon_F = \frac{h^2}{2m} \left(\frac{3N}{8\pi V} \right)^{2/3} \quad (\text{i})$$

where N/V is the density of free electrons.

Electron energy distribution

No. of electrons in the electron gas having energy between ϵ and $\epsilon + d\epsilon$ is

$$n(\epsilon)d\epsilon = g(\epsilon)f(\epsilon)d\epsilon$$

$$n(\varepsilon)d\varepsilon = \frac{8\sqrt{2}\pi V m^{3/2}}{h^3} \frac{\varepsilon^{3/2} \sqrt{\varepsilon}}{e^{(\varepsilon - \varepsilon_F)/kT} + 1} d\varepsilon$$

using (i)

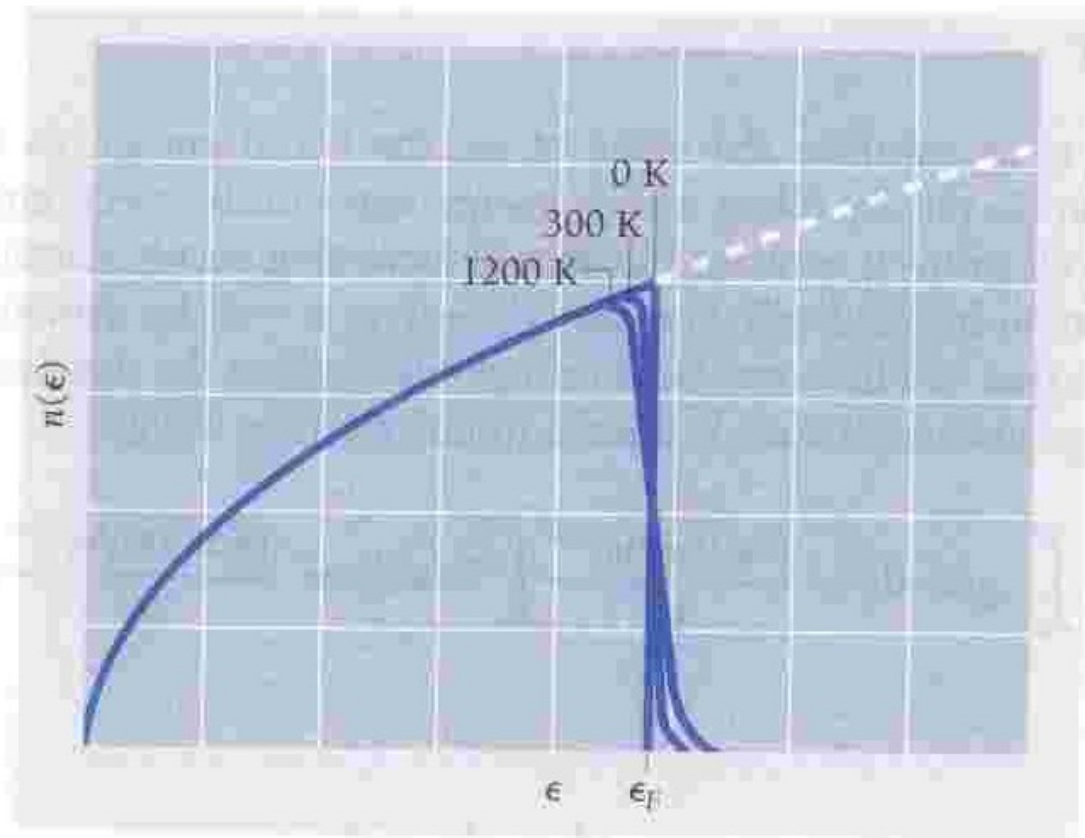
$$n(\varepsilon)d\varepsilon = \left(\frac{3N}{2} \right) \frac{\varepsilon^{-3/2} \sqrt{\varepsilon}}{e^{(\varepsilon - \varepsilon_F)/kT} + 1} d\varepsilon$$

This is electron energy distribution formula, according to which distribution of electrons can be found at different temperatures.

When a metal is heated then only those electrons which are near the top of the fermi level (kT of the Fermi energy) are excited to higher vacant energy states.

$$kT = 0.0025 \text{ eV at } 300\text{K}$$

$$kT = 0.043 \text{ eV at } 500\text{K}$$



The electrons in lower energy states cannot absorb more energy because the states above them are already filled.

This is why the free electrons contribution in specific heat is negligible even at high T .

Average electron energy at 0K

Total energy at 0K is

$$E_o = \int_0^{\epsilon_F} \epsilon n(\epsilon) d\epsilon$$

Since at 0K all electrons have energy less than or equal to ϵ_F

$$e^{(\epsilon - \epsilon_F)/kT} = e^{-\infty} = 0$$

$$E_o = \frac{3N}{2} \epsilon_F^{-3/2} \int_0^{\epsilon_F} \epsilon^{3/2} d\epsilon = \frac{3N}{5} \epsilon_F$$

Then average electron energy

$$\left(\bar{\epsilon}_o \right) = \frac{E_o}{N} = \frac{3}{5} \epsilon_F \quad \text{at 0K.}$$